



Conversational User Interfaces

Recap & Last Quiz

Lecture Quiz

In your own words, explain the **difference of zero-shot and few-shot prompting**. How would it impact a user if a system only works well with few-shot prompting?

“**Zero-shot prompting** means giving a language model a task without providing any examples. You **just describe what you want**” → **Instruction only**

“**Few-shot prompting** means including a **small number of input-output examples** in the prompt before the actual task”

Lecture Quiz

Choose and answer one:

- **Pain points of prompting in your own experience:** Think about your use of prompting UIs (e.g. ChatGPT and similar chatbots or other systems you might have used). **Is there something that you find annoying or less-than-ideal** about it? Explain that "pain point" and suggest an improvement.
- **Potential of prompting in software you use:** Think about software that you use (and that does not involve prompting yet or not in the way you would like). **Describe a potential UI feature idea** for that software that you expect might improve that software (for you or in general) by involving a prompting-based interaction or functionality.

Lecture Quiz

Potential for a new prompt-based feature

“I would add a prompting feature to a **music app** like Spotify.

Users could type prompts such as:
“Play relaxing music for studying”

The AI could then **generate personalized playlists** automatically.
This would make searching faster, more natural, and more personalized than only using buttons or categories.”

Note: What I liked about this idea is that it uses the potential of natural language prompts to express *vague* ideas and matches this with the fuzziness (i.e. “vibe”) of musical preferences and contexts.

In other words, a UI making it easy for users to stay vague with their input can be a problem in other use cases of prompting (as some of you mentioned in your answers for the “pain points” question) but here it seems to fit.

Lecture Quiz

Pain point + feature idea to address them

“A common pain point in Chatgpt is the **lack of memory continuity across sessions**. You often have to re-explain your context every new conversation, which breaks workflow.

A simple improvement would be a persistent user profile **panel where you can store preferences, roles, and recurring context** that the system automatically includes without you having to retype it every time.”

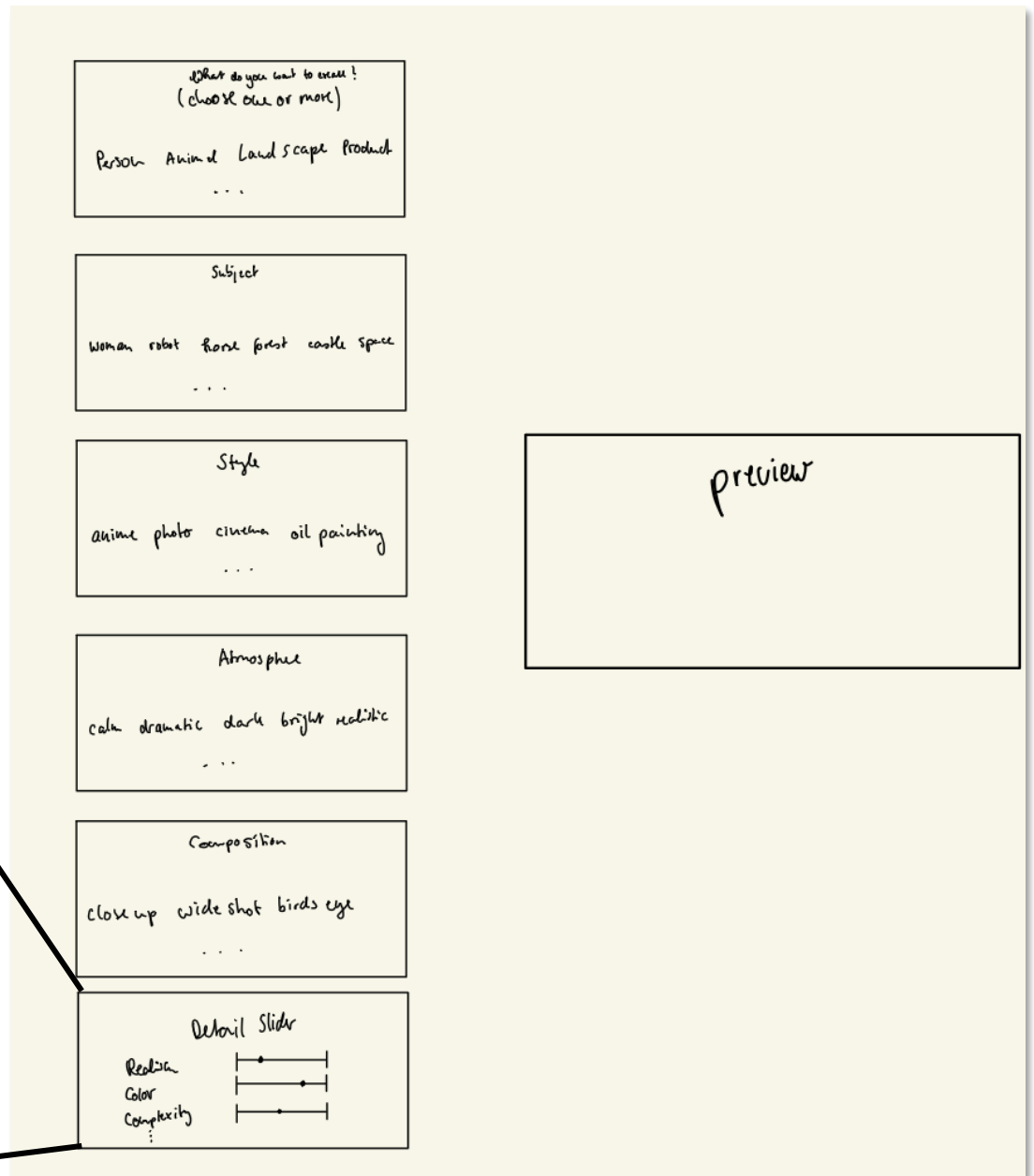
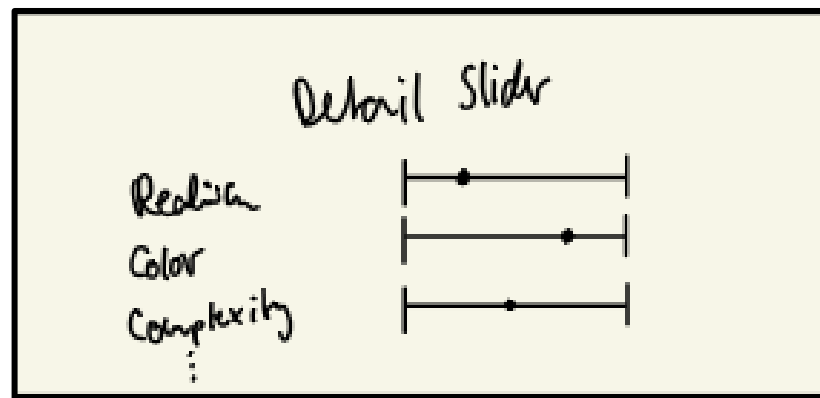
Note: What I liked about this idea is that it's not just about storing past conversations but rather exposing this to the user in an interactive way in the UI (“panel where *you* can...”).

- If we see this info in the panel as a user model, then this is known as a “*scrutable*” and/or “*editable*” user model.
- If we see this info in the panel as additional input, then the panel facilitates more “*structured prompting*”, by splitting the overall prompt input into the typical prompt (instruction in a chat) and the specified context (in the panel).

Lecture Quiz

UI ideas for instructing a text-to-image generation system without entering text

Splits the input UI into multiple inputs for different aspects, some of which do not need text input:

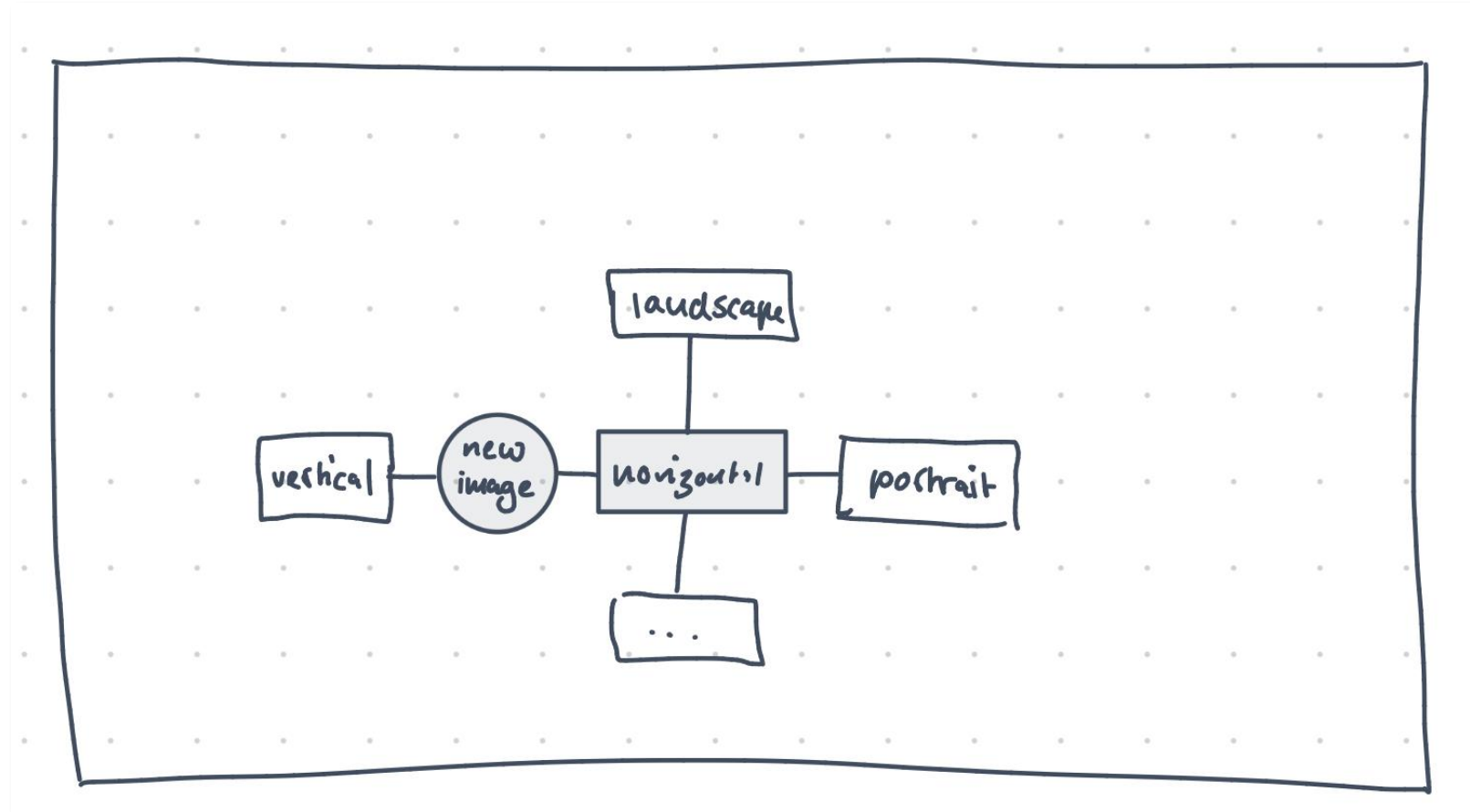


Lecture Quiz

UI ideas for instructing a text-to-image generation system without entering text

“My Idea is a text-to-image generation system that works similarly to a **mind map**. Whenever the user is clicking a **bubble**, it decides another **aspect** of the image and new bubbles will appear with further details.

One bubble is always left empty so that the user can, if necessary, add their own aspects.”

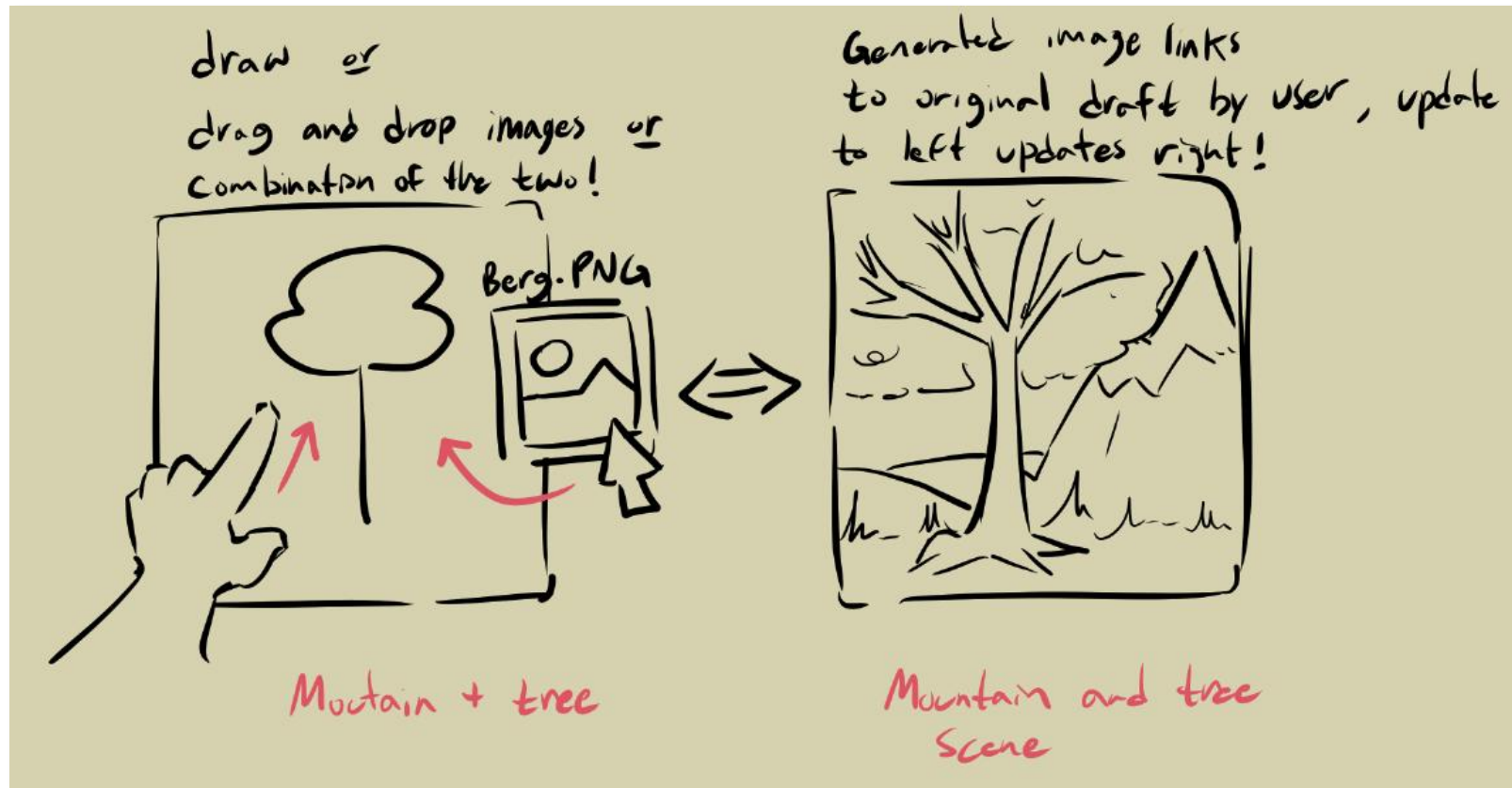


Lecture Quiz

UI ideas for instructing a text-to-image generation system without entering text

Composer view:

Drag & drop of images to compose a new image/scene



Output view:

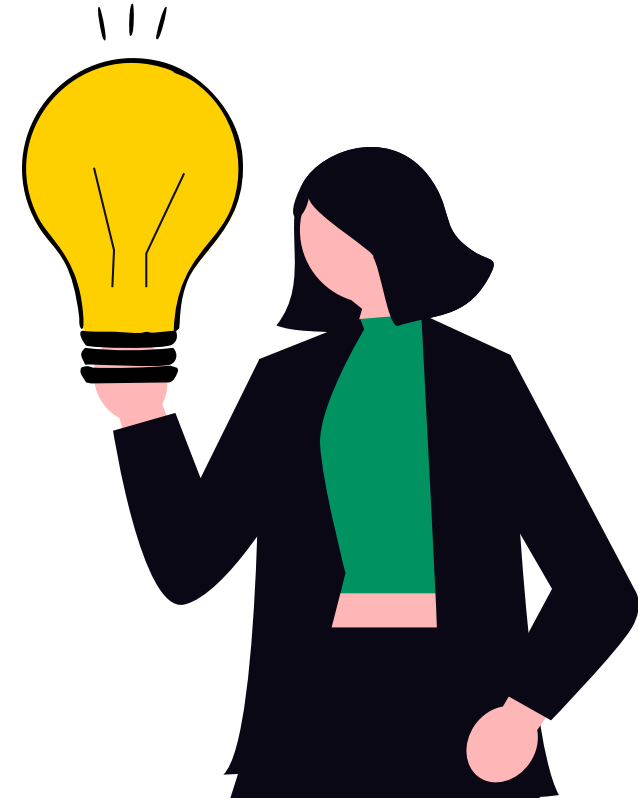
Show the generated image next to the composed input, updated live

This lecture

Learning goals

After this session, you should be able to:

- ✓ **List** important design factors for CUIs and **explain** each of them with an example, incl. pros and cons of specific choices
- ✓ **Sketch** a dialog flow for a given use case, which includes the discussed elements, where useful
- ✓ **Analyse** a given use case to give a recommendation for how to use (or rather not use) a conversational UI (or UI feature) here
- ✓ **Explain** the “media equation” in your own words and its impact on what we need to consider in the design of CUIs



Intro

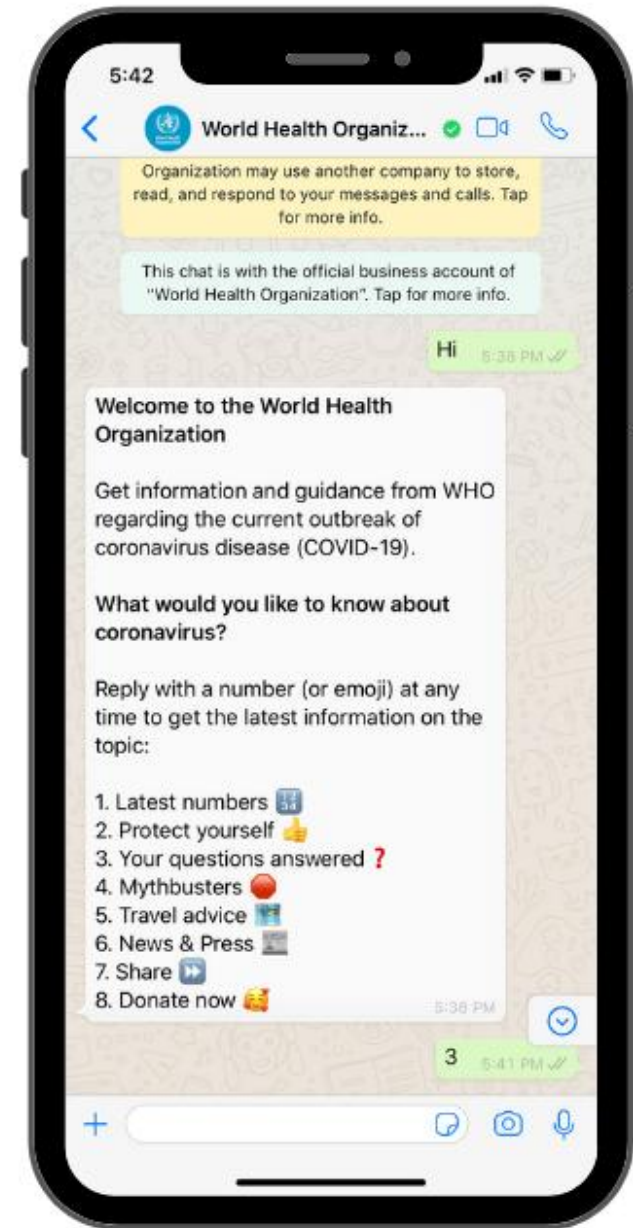
Example

Chatbot by the World Health Organization

- Provided covid19 infos via chat (WhatsApp)
- What are the benefits of this approach?
Why not just a website?

<https://www.who.int/news-room/feature-stories/detail/who-health-alert-brings-covid-19-facts-to-billions-via-whatsapp>

<https://www.whatsapp.com/coronavirus/who>



Activity

Conversational user interfaces

Why conversational UIs?

What are their **benefits** and **challenges**?

Think about both **voice/speech** interaction
and **text-based** interaction (e.g. chatbot).

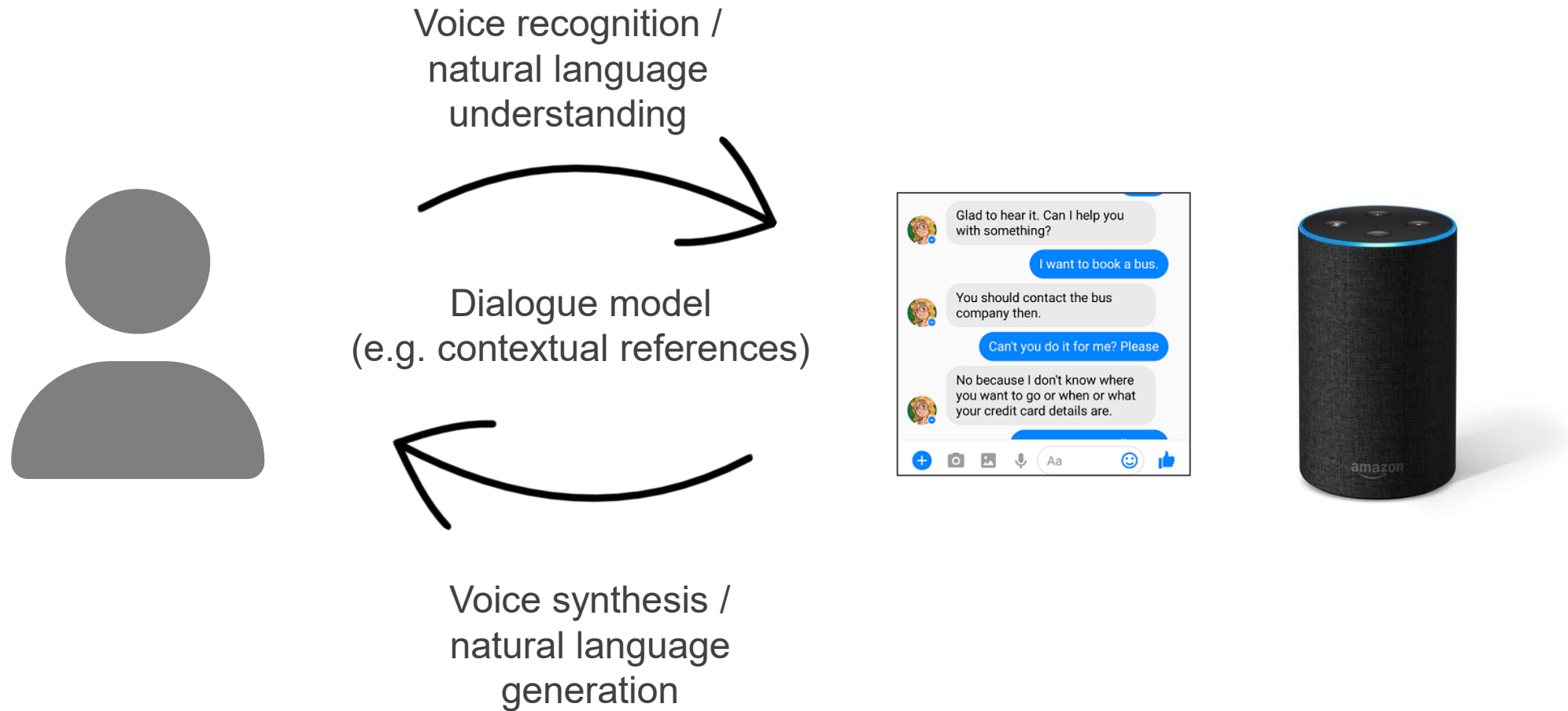


Motivation for CUIs

Conversations...

- can feel natural (known from human to human)
- support hands-free input (via voice)
- don't require a graphical display (via voice)
- keep context/history
- can be *about* something (e.g. main task + conversation about it in parallel)
- accommodate negotiation and clarification
- go well with prompting (see recent hype)
- communicate personality (see movie dialogues)
- ...

Involved tasks & technologies



Technical approaches

Machine Learning vs Rule-based

- Used to be **rule-based**
(and might still be in legacy systems in production)
- Now & envisioned: mostly **prompting LLMs**
- Except for one? kuki

“Kuki holds a world-record for **winning the Loebner Prize, an annual Turing Test Competition, five times**. Kuki is implemented primarily using an open standard, **rule-based scripting language** called Artificial Intelligence Markup Language (AIML), which entails **hand-authoring chatbot replies** in response to an analysis of incoming user input data with a **blend of statistical models, machine learning, and manual review/tagging**. This hybrid methodology has numerous key advantages including [...] almost zero response latency, imperviousness to toxicity and corruptibly, and general suitability to brand-appropriate production use cases.”

<https://www.kuki.ai/research>



How do people use CUIs?

Studies on voice assistants at home, over the last years

From **task-oriented** (what do people ask these devices?)...

“Music, Search, and IoT: How People (Really) Use Voice Assistants”

Music	pause, spotify, pandora, music, skip, next, song, stop, play, alexa	stop, play, skip, shuffle, song, lullaby, music, sing, radio, pause	shuffle songs by dropping young
Search	many, list, song, echo, left, tell, much, time, alexa	echo, time, find, state, white, know, thing, series, score, twenty	alexa how many hours are in a year, what states have the death penalty
IoT	ten, set, kitchen, percent, bedroom, living, room, alexa, light, turn,	bedside, turn, door, kitchen, set, light, lamp, percent, room, bed	echo bedside off, echo turn on kitchen light
Volume	eight, ten, seven, four, three, six, five, turn, alexa, volume,	echo, turn, volume, three, level	alexa turn the volume to six
Conversational	hello, play, okay, thank, morning, shut up, hey, night, good, alexa	good, morning, thank, okay, shut up, series, bedtime, story, hello, robot	tell bedtime story to [name redacted], alexa who is your favorite robot
Timer	twenty, thirty, ten, left, five, much, set, minute, alexa, time	timer, remind, add, restart, many, count, delete	alexa how many timers do i have set, alexa delete timer

... [\[Ammari et al., 2019\]](#)

... to **situated** (how do people integrate these into their lives).

“Finding a Home for Voice Assistants: A Domestication Calculus Across Three Years and Thirty Households”

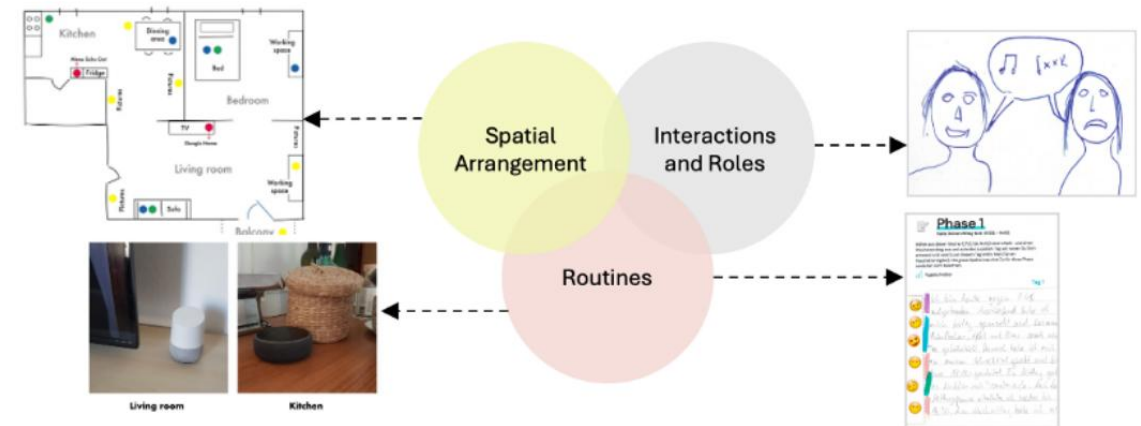


Figure 1: The domestication calculus of voice assistants across three aspects — spatial arrangement, interactions and roles, and routines. The diagram illustrates how we traced domestication through diverse materials collected in a three-year field study, including home maps, photographs, diary snapshots, sketches and interviews with participants.

[\[Alizadeh et al., 2026\]](#)

CUI Design

Design factors
to consider for conversational user interfaces

Modality

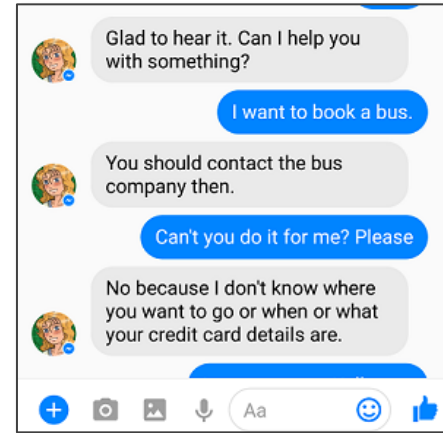
Conversation via speech/voice vs text



Choice usually depends on the use-case and context; rarely a “free” design choice

Speech / voice

- + Hands-free (e.g. in the car)
 - + Paraverbal cues
-
- Status/progress hard to display
 - Time pressure
 - Not suitable in some contexts (e.g. noise, social)



Also possible “mix”:
Text input via speech/voice
+ output via text

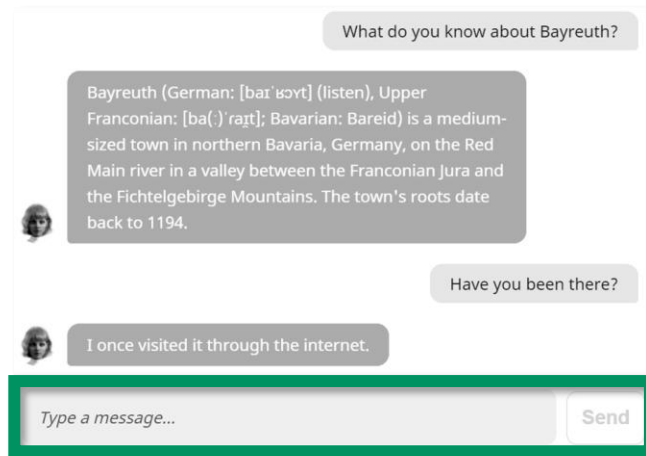
→ keeps visual requirements
(screen + looking at screen)

Text

- + Integration with GUI elements & visual content
 - + Status visibility easier to display
-
- Requires screen
 - Requires text input method
 - Requires visual attention

Conversational text input

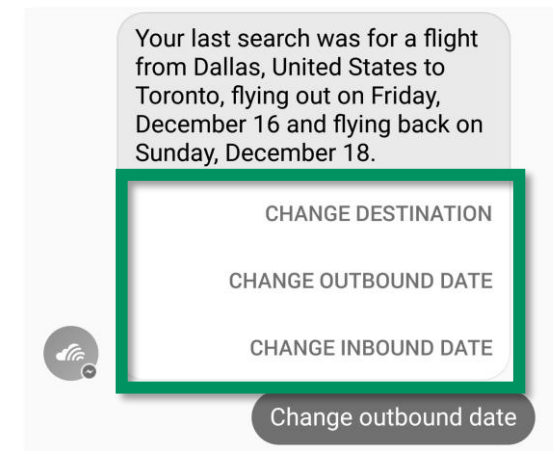
Free text vs fixed options



Choose depending on requirements of emulating a “real” conversation and/or functional flexibility

Free text

- + Feels more like a “real” conversation
 - + Functional flexibility (e.g. ask anything)
 - + Free user expression (e.g. write in your style)
-
- No guardrails, potentially unwanted in/output, edge cases
 - Does not communicate (limits of) functionality



Fixed options

- + Better control (for designer / developer)
 - + User guidance, communicates functionality, predictable
 - + Lower input effort for user
-
- Feels less like a “real” conversation
 - Limited scope, which might miss what user wants/needs

Activity

Transferring UI design principles from GUIs to CUIs

Here are 4 typical GUI design principles. Discuss with your neighbour(s):

Could you transfer these principles from GUIs to CUIs (voice and/or text-based) or not?

If so, how?



1. Feedback

How can CUIs realize useful feedback?



2. Consistency

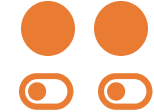
How might we realise consistency in CUIs?

Or: What should be kept consistent?



3. Metaphors

Can CUIs use metaphors?



4. Spatial mappings

Can CUIs use such mappings?

Additional (visual) elements

Associated with the system's output

If you're looking for a specific airport just reply back with the airport codes you're looking for like "I want DFW to AZA only".

Depart: Tue, Feb 21	American	
DFW 8:25 pm	2h39m nonstop	PHX 10:04 pm
Return: Tue, Feb 28	American	
PHX 10:52 pm	2h12m nonstop	DFW 2:04 am
\$95		\$95
#1 Cheapest departing Dallas, TX after 3pm, nonstop		#2 C depa

Key benefit for text-based conversations on a screen, so think about useful chat-integrated UI elements for your use-case



@kuki_ai

👋🌐 I'm Kuki, your AI 🐱 BFF ❤️

Can shape user expectations even before the first conversation



Other GUI elements

e.g. status, progress bar, links, tables, images, ...

Surrounding presentation

e.g. avatar, "profile" for the chatbot, ...

Physical device

e.g. device design, form factor, lights, ...

Output design

Design dimensions tied to the conversation text + voice

Linguistic features (used words)

Key way of shaping perception of the “personality” of a chatbot; should fit use-case / context

Paraverbal features (used voice)

Word choice

Sentence length

Level of formality

Puns

Cultural proverbs

Slang

Irony

Prosody

Dialect

Tone of voice

Intonation

Gender

Some guidelines

That work for the design of conversational UIs



Tell users **what they can do**

“You can ask for today’s weather or a weekly forecast.”



... and **where they are**

“Today’s forecast is sunny and dry”
(rather than just „sunny and dry.”)



Prefer **examples over instructions**

e.g. in the greeting or a help feature



Limit the amount of **information**

e.g. max 3 options per interaction
(users need to keep these in short-term memory)



Use **visual feedback**

if possible
(e.g. light on physical device; indicators for on-screen chatbots)

Conversation Design

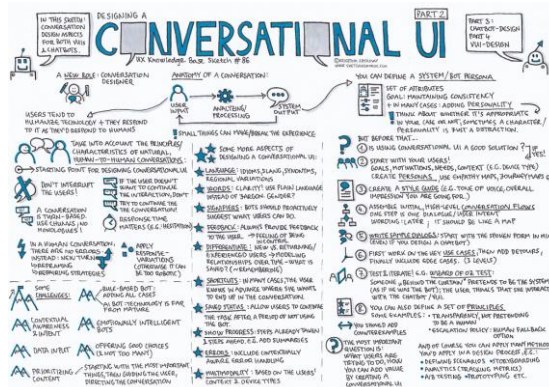
Designing and implementing the actual conversation,
e.g. its structure and flow

Design process

For conversational UIs

We'll try out this step 4 next!

Check the other steps and the summary graphic ([link](#)) with your team if your project has a CUI!



<https://uxknowledgebase.com/conversational-ui-designing-part-2-52bad56f005c>

1. IS USING CONVERSATIONAL UI A GOOD SOLUTION? IF YES!
 2. START WITH YOUR USERS!
GOALS, MOTIVATIONS, NEEDS, CONTEXT (E.G. DEVICE TYPE)
CREATE PERSONAS, USE EMPATHY MAPS, JOURNEYMAPS ETC.
 3. CREATE A STYLE GUIDE (E.G. TONE OF VOICE, OVERALL IMPRESSION YOU ARE GOING FOR.)
 4. ASSEMBLE INITIAL, HIGH-LEVEL CONVERSATION FLOWS
ONE STEP IS ONE DIALOGUE / USER INTENT
WORDING: LATER; IT SHOULD BE LIKE A MAP
 5. WRITE SAMPLE DIALOGS: START WITH THE SPOKEN FORM IN MIND (EVEN IF YOU DESIGN A CHATBOT)
 6. FIRST WORK ON THE KEY USE CASES, THEN ADD DETOURS, FINALLY INCLUDE EDGE CASES. (3 LEVELS)
 7. TEST & ITERATE! E.G. WIZARD OF OZ TEST:
SOMEONE „BEHIND THE CURTAIN“ PRETENDS TO BE THE SYSTEM (AS IF HE WAS THE BOT); THE USER THINKS THAT SHE INTERACTS WITH THE CHATBOT / VUI.
 8. YOU CAN ALSO DEFINE A SET OF PRINCIPLES
SOME EXAMPLES:
 - TRANSPARENCY, NOT PRETENDING TO BE A HUMAN
 - ESCALATION POLICY: HUMAN FALLBACK OPTION
- ↔ YOU SHOULD ADD COUNTEREXAMPLES
- THE MOST IMPORTANT QUESTION IS:
WHAT USERS ARE TRYING TO DO, HOW YOU CAN ADD VALUE BY CREATING A CONVERSATIONAL UI
- AND OF COURSE YOU CAN APPLY MANY METHODS
YOU'D APPLY IN A DESIGN PROCESS, E.G.:
- DEFINING SCENARIOS
 - STORYBOARDING
 - ANALYTICS (TRACKING METRICS)
 - A/B TESTING
 - PROTOTYPING ETC.

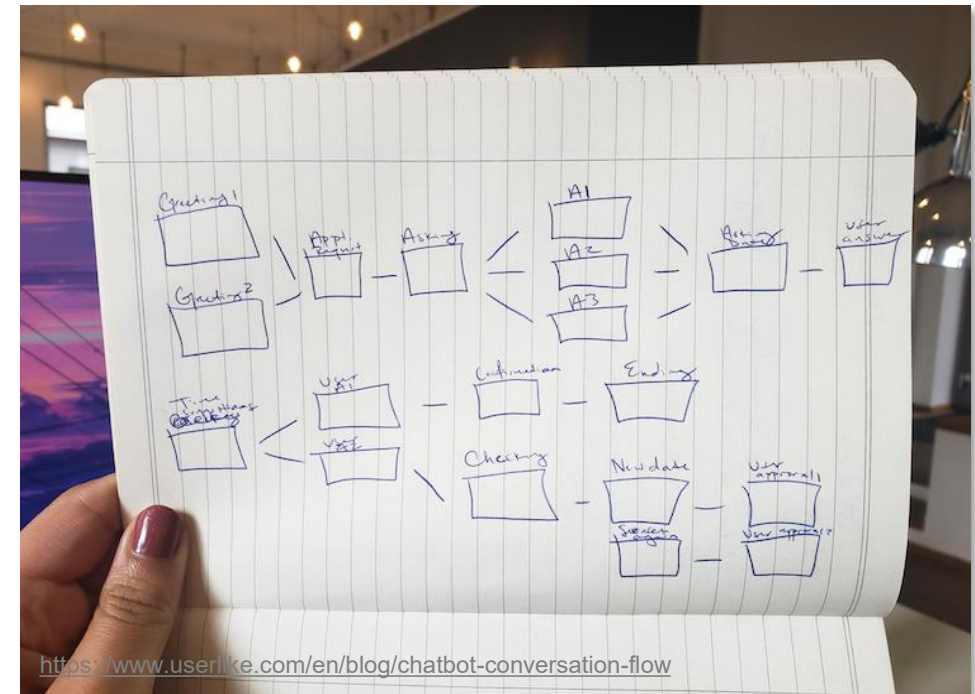
Activity

Conversation Design

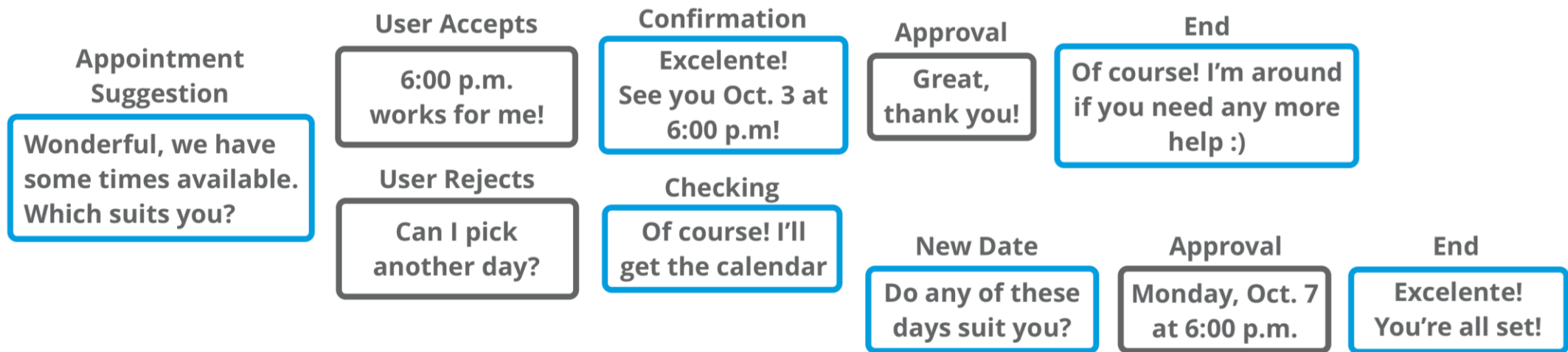
Scenario: You develop a chatbot for **scheduling appointments at a local business** (e.g. hair stylist, bookshop, doctor).

Task:

1. Create a **conversation flow diagram** for this.
2. Can you identify conversation **elements** that could be generally useful, beyond your specific scenario?



Example

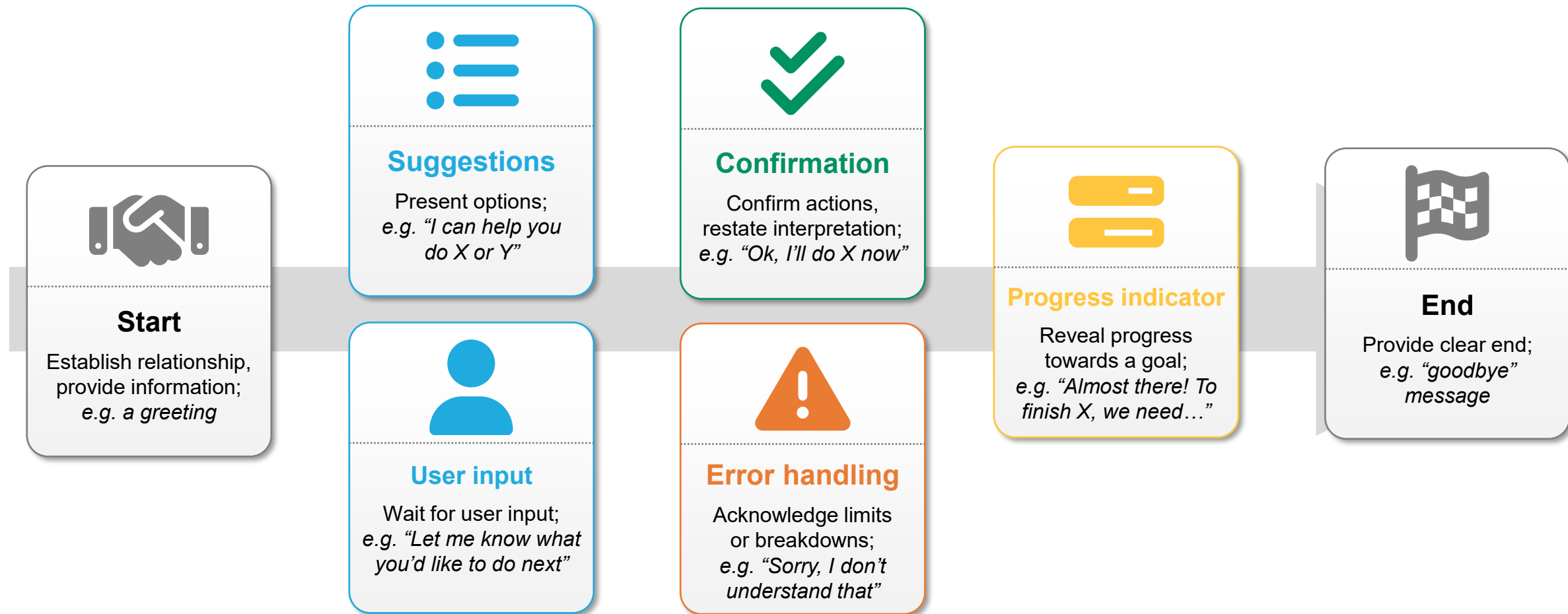


<https://www.userlike.com/en/blog/chatbot-conversation-flow>

Conversation elements

Typical elements for your conversation flow

The “middle part” here is typically repeated;
e.g. for each required piece of information
or “step” in a procedure flow

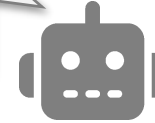


Breakdowns

Some “repair strategies”

Repeat

I don't quite understand. How can I help you with your banking needs today?

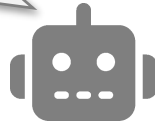


Note: “not understanding” is not the only breakdown. Others include e.g. requests beyond what the bot can do (or know) or beyond what it is allowed to do.

Example:

Bot doesn't understand this

Hello, I'm your digital assistant. How can I help you with your banking needs today?



Hi, I'd like to add my daughter to my card.



Explain

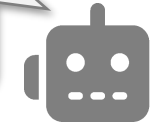
Hi, I'd like to **add my daughter** to my card.

Sorry, I don't understand. I've **highlighted** the part I'm unable to process. Could you rephrase please?



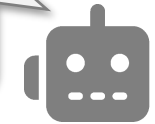
Give options

I'm not quite sure if I understood. Which of these are you trying to do today: 1) Add ... 2) ... 3) ... 4) None of the above?



Confirm

Sounds like you want me to add a credit card to your account. Is that right?



Defer

Sorry, I don't understand. I'll refer you to a human agent who will respond in the next few minutes.



Overall:

- Acknowledge breakdowns
- Assist user repair
- Proactively suggest solutions

[Ashktorab et al., 2019]

Implementing conversations

Tools & approaches

- **Dedicated frameworks, e.g.:**
 - Rasa: <https://legacy-docs-oss.rasa.com/docs/rasa/>
 - Dialogflow: <https://cloud.google.com/products/conversational-agents>
 - IBM AI agents (also automation): <https://www.ibm.com/ai-agents>
 - The recent “agentic” frameworks (if they include user-facing CUIs)
- **Prompting-based approach** (see next slide), e.g.:
 - Core component is an **LLM** (or API to an LLM)
 - Write custom **prompts for intent recognition**
(e.g. to classify a user request into X specified requests,
e.g. “weather today”, “forecast next day”, “forecast week”)
 - Write custom **prompts to generate the output** for each intent
(e.g. to verbalise the respective forecast as a short message)

Conversational Agents

Build conversational AI agents that provide proactive, personalized self service and natural, human-like voices for highly engaging customer experiences.

[Go to console](#)

[Request a free consultation](#)

- ✓ Support rich, intuitive customer conversations, powered by Google's leading generative AI

✓ En [Welcome to Rasa](#)

har

✓ On

virt

✓ Cre

✓ Par

Welcome to the Rasa Docs

The Rasa Platform helps teams build, test, deploy, and analyze AI assistants at scale. Whether launching your first assistant or scaling enterprise AI, Rasa is set up to adapt and grow with you.

Quickstart

Start building with Rasa for free.

Build without code

Start a free 7-day trial of Rasa Studio.

Learn More

Grow your understanding of Rasa concepts.

BENEFITS

Trans

Core Concepts

Dive deeper into the concepts behind Rasa.

Guides

Have a usecase in mind? Check out our Guides for how Rasa helps you build assistants.

Team Resources

Jumpstart your assistant and setup your team for success with some best practices.

Dive into the Products

Find detailed guidance for each step of building assistants in our product docs.

Pro

The powerful pro-code interface for building assistants.

Studio

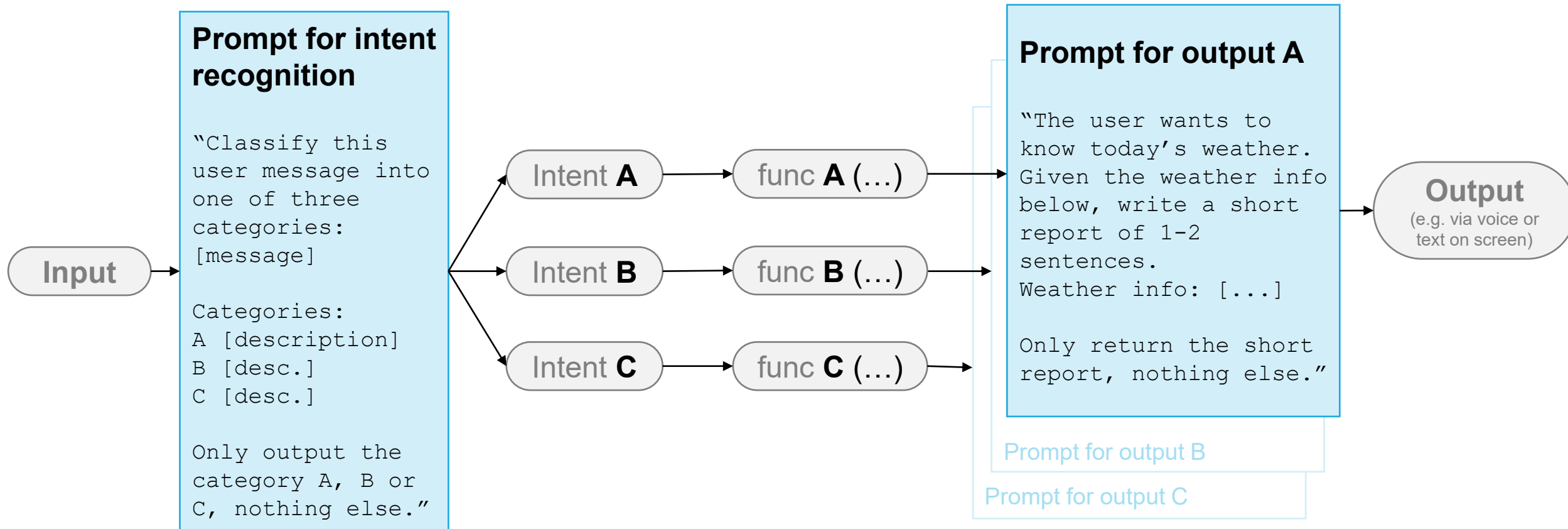
The intuitive no-code interface for building assistants.

Last updated on May 30, 2025

Next
[What You Can Build with Rasa](#)

Prompting-based approach

E.g. for prototyping a conversational agent with specified functionality



Prompting-based approach

... with LLM function calling

“Function calling”

- Describe what a function does in natural language
- This allows an LLM to decide itself when to call which function
- The function itself can then be anything (e.g. a hardcoded formula, retrieving a json from a weather API, etc.)

Function calling example with get_weather function

python ↕

```
1 from openai import OpenAI
2
3 client = OpenAI()
4
5 tools = [{
6     "type": "function",
7     "name": "get_weather",
8     "description": "Get current temperature for a given location.",
9     "parameters": {
10         "type": "object",
11         "properties": {
12             "location": {
13                 "type": "string",
14                 "description": "City and country e.g. Bogotá, Colombia"
15             }
16         },
17         "required": [
18             "location"
19         ],
20         "additionalProperties": False
21     }
22 }]
23
24 response = client.responses.create(
25     model="gpt-4.1",
26     input=[{"role": "user", "content": "What is the weather like in Paris today?"}],
27     tools=tools
28 )
29
30 print(response.output)
```

<https://platform.openai.com/docs/guides/function-calling?api-mode=responses>

Prompting-based approach

... with a system prompt

Instead of the hand-built flow on the previous slides, we could just put a **description of the (high-level) behaviour into the system prompt** for your LLM.

What might be the **pros & cons** here?
(also see the last session on prompting)



See the “herding cats” problem / paper from the last session!

[Zamfirescu-Pereira et al., DIS 2023]

System prompt

“You’re a helpful assistant for scheduling appointments for our business. Adhere to these rules:

- Always be polite.
- Don’t respond to questions beyond appointment scheduling.
- Check our internal calendar tool before suggesting any appointment slot.
- ... ”

*Note: such system prompts can quickly end up with lots of “if X then Y” statements – just in natural language rather than code
→ new tools for developers to handle this?*

Conversational agents with personality

Can computers have “personality”?

How does that impact interaction?

How might we consider that in the design of CUIs?

The „Media Equation“

People tend to **assign human characteristics to media** (incl. computers), treating them as social actors

Examples:

- Giving a name to your vacuum cleaner, laptop, car, etc.
- Being polite to a machine (e.g. saying „thank you“ to a voice assistant or chatbot)
- Antropomorphising technology (e.g. „my printer has a bad day“)
- „Feeling“ for ChatGPT etc. if it generates text that expresses mood, emotions, opinions, etc.

[Reeves & Nass, 1996]

For CUIs:

- Conversational agents are seen as such actors
- People assign them a personality (whether you’ve designed it or not)



Personality and Interaction



Activity

Your impressions of a conversational agent

Here are some CA products.
Discuss with your neighbour(s):

If you've used multiple CAs: **Which one is the most...**
Or: **To what extent is the one you've used...**

- Funny?
- Respectful?
- Capable?
- Vague?
- Lazy?
- Excitable?



ChatGPT

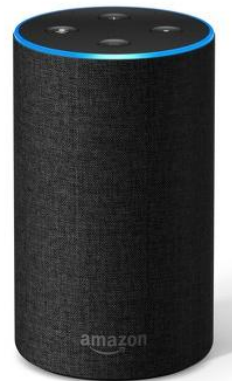


Google Assistant

 Claude



Hey Siri



Modelling personality

For humans

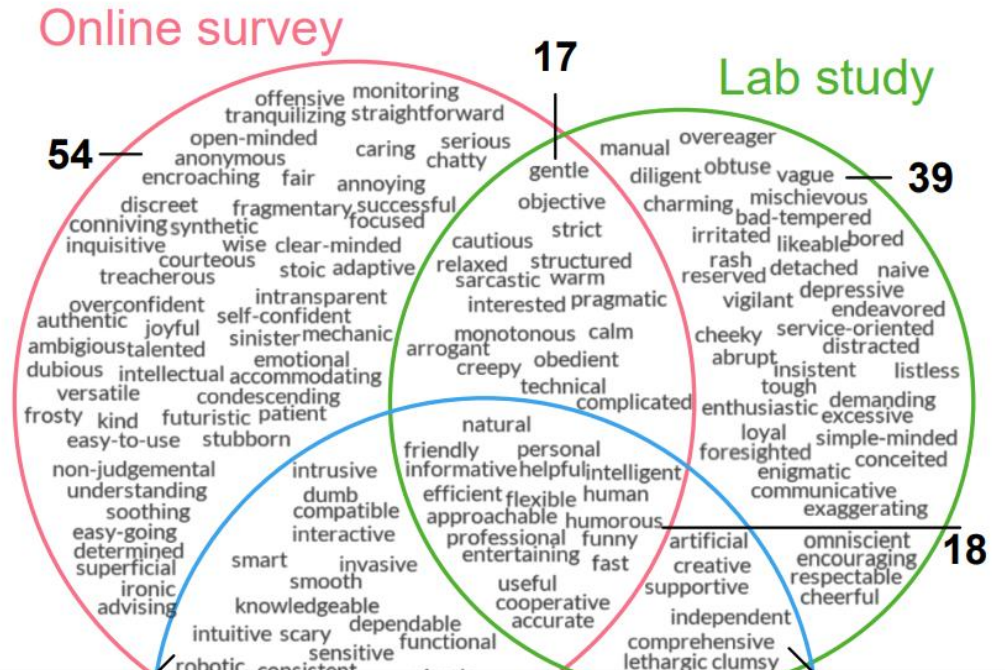
- Personality is present in **thinking, feeling, behaving**
- Considered **stable** over life and situations (e.g. in contrast to emotions)
- **Big Five** personality model („OCEAN“)
 - Openness
 - Conscientiousness
 - Extraversion
 - Agreeableness
 - Neuroticism

[McCrae & Costa, 2008]

Modelling personality

For conversational agents (CAs)

- Idea: Dedicated personality model for CAs
- Psycholexical approach:
 - Collect adjectives
 - Let people rate CAs using these
 - Run factor analysis on the ratings (results see next slide)



	untypical	rather untypical	rather typical	typical
self-disciplined	☐	☐	☐	☐
adorable	☐	☐	☐	☐
ambitious	☐	☐	☐	☐
obtuse	☐	☐	☐	☐

Text analysis

[Völkel et al., 2020]

Results

Personality dimensions for CAs

- Found 10 dimensions
- Need to be confirmed, refined over time
- Might change as technology and people's experiences and expectations change
- Full descriptions see [paper](#)

Factor / Label	Top 20 descriptors (by factor loadings)
1 Confrontational	abusive (.71), negligent (.70), deceitful (.69), cruel (.68), distrustful (.67), combative (.65), offensive (.65), incomprehensible (.64), stingy (.62), messy (.60), encroaching (.59), scornful (.58), dubious (.58), irritable (.58), manipulative (.58), explosive (.56), clumsy (.55), condescending (.54), clunky (.54), vindictive (.53)
2 Dysfunctional	reckless (.71), lazy (.71), irritated (.70), fearful (.69), bitter (.65), moody (.65), crazy (.63), unmanageable (.62), prejudiced (.61), treacherous (.61), ignorant (.59), conceited (.59), dead (.58), inoperable (.57), forgetful (.57), bad-tempered (.55), impetuous (.55), listless (.52), stubborn (.51), egocentric (.50)
3 Serviceable	informative (.56), functional (.56), capable (.56), accurate (.55), knowledgeable (.55), convenient (.54), thorough (.52), responsive (.52), productive (.52), consistent (.50), useful (.50), user-friendly (.50), interactive (.50), adaptive (.50), helpful (.50), communicative (.49), organized (.49), dependable (.48), intelligent (.47), comprehensive (.47)
4 Unstable	nervous (.56), depressive (.54), rude (.54), bigoted (.52), grumpy (.51), jealous (.49), forceful (.49), anxious (.48), gruff (.47), easy-to-use (-.47), frosty (.47), fretful (.45), tempestuous (.45), dangerous (.45), sloppy (.45), faultfinding (.45), dumb (.45), troublesome (.43), rambunctious (.41), temperamental (.41)
5 Approachable	peaceful (.64), easy-going (.62), gentle (.61), relaxed (.61), fair (.57), clear-minded (.56), respectful (.53), calm (.53), humble (.51), respectable (.48), casual (.48), courteous (.47), sincere (.46), understanding (.44), ethical (.44), loyal (.43), open-minded (.43), principled (.42), simplistic (.42), determined (.39)
6 Social-Entertaining	humorous (.69), playful (.57), funny (.66), joyful (.52), charming (.51), entertaining (.51), cheerful (.50), merry (.49), happy-go-lucky (.49), cheeky (.48), expressive (.45), chatty (.45), affectionate (.45), happy (.45), excited (.44), enthusiastic (.44), social (.42), adorable (.41), warm (.39), encouraging (.38)
7 Social-Inclined	agreeable (.51), willing (.46), likeable (.43), kind (.42), trustful (.41), decent (.40), modest (.40), flexible (.40), interested (.38), realistic (.38), conversational (.38), friendly (.37), self-disciplined (.37), inquisitive (.35), patient (.34), soothing (.32), endeavored (.32)
8 Social-Assisting	pragmatic (.61), fastidious (.58), scrupulous (.57), genial (.54), diplomatic (.46), omniscient (.45), vigilant (.44), vigorous (.44), benevolent (.43), dignified (.43), amiable (.41), stoic (.38), conscientious (.36), discreet (.36), deliberate (.35), meticulous (.33), lenient (.33), zestful (.32), foresighted (.32), restrained (.31)
9 Self-Conscious	independent (.45), assertive (.45), competitive (.43), brave (.42), creative (.42), deep (.41), selective (.41), proud (.39), excitable (.39), artistic (.37), ambitious (.37), introspective (.36), powerful (.35), individualistic (.35), self-indulgent (.35), insistent (.35), extravagant (.34), crafty (.34), contemplative (.34), daring (.34)
10 Artificial	synthetic (.50), robotic (.49), intrusive (.48), artificial (.47), odd (.45), invasive (.44), gimmicky (.44), abrupt (.43), superficial (.43), monotonous (.43), fake (.42), simple-minded (.41), mechanic (.41), vague (.41), passionless (.40), digital (.39), electronic (.39), monitoring (.39), annoying (.36), detached (.35)

Table 3. Overview of the factors obtained in our exploratory factor analysis, with top 20 descriptors (and their factor loadings). The factor labels suggested here are based on the authors' interpretation of all descriptors per factor with loadings of at least .3.

[Völkel et al., 2020]

Designing personality

How can we deliberately design CA personality?

■ What to design?

- Presentation: avatar, device, surrounding presentation, etc.
- Conversation:
 - Voice attributes (e.g. pitch, speed)
 - What is said and how: Word choice, sentence length, slang, level of formality etc.
 - Conversational structure: Initiative / proactiveness, questions, clarifications, etc.

■ How to design?

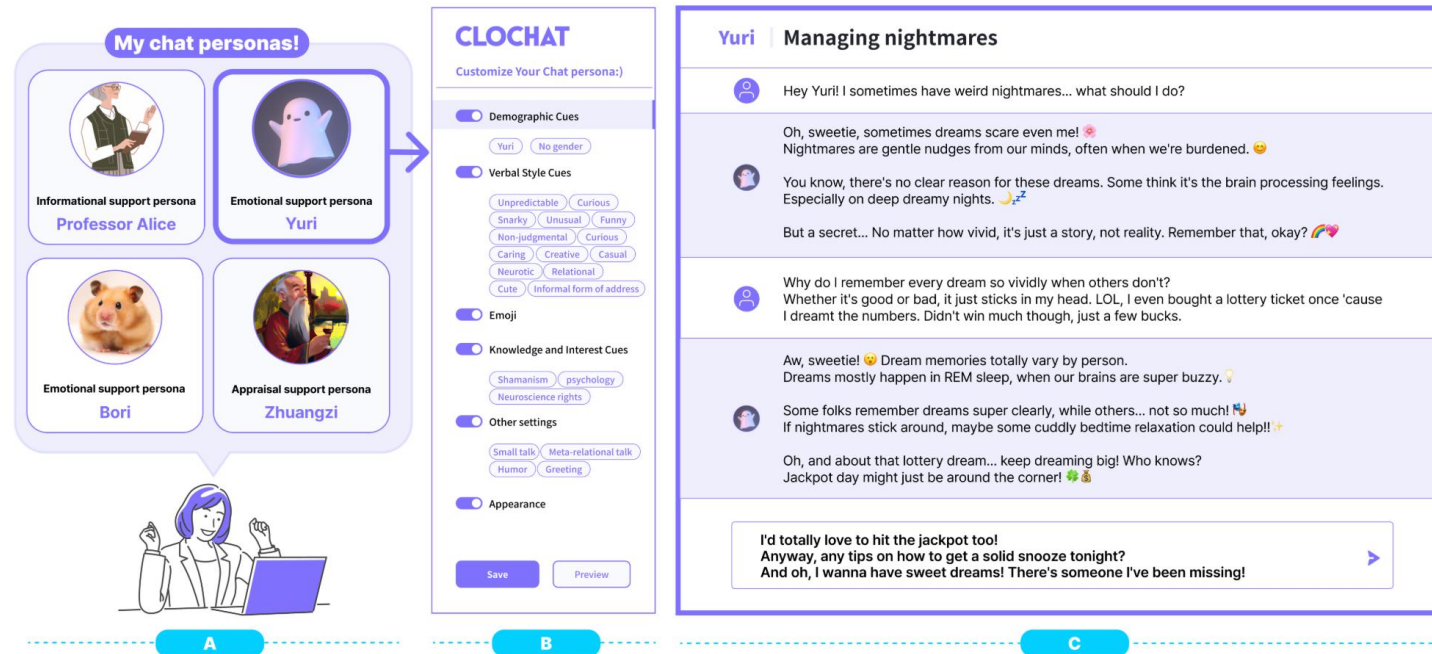
- Top-down: Based on concepts from prior work/literature (e.g. extraverted CA might talk faster, louder, more)
- Bottom-up: Based on user feedback and people (e.g. enactment-based dialogue design – i.e. ask actors to play a conversation with a certain personality)



[Völkel et al., 2021]

System example

Of considering personality for a conversational agent



Key idea:

Giving **users** the opportunity and UI controls to create or **customize** the **personality** of their agent (through various concepts, incl. personality dimensions from Psychology).

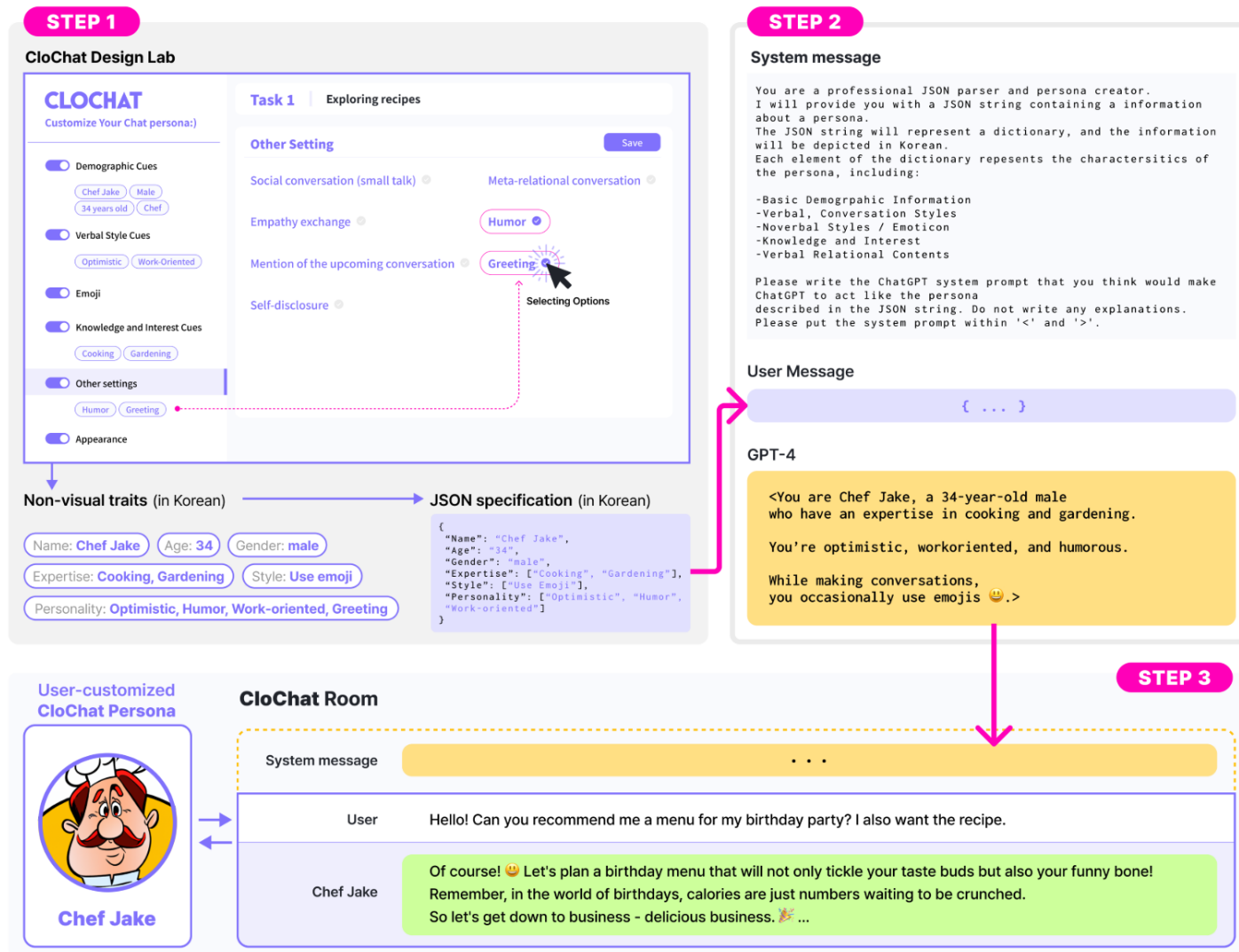
Figure 1: CloChat supports users in creating and interacting with bespoke agent personas. Using CloChat, users can materialize the personas in their minds (A) by interactively customizing their traits (B). Based on user customization, CloChat automatically generates the agent persona. Users can then freely converse with the created agent personas (C). Our research showed that agent personas customized with CloChat (1) substantially enhanced the participants' conversational experiences, (2) significantly increased the diversity of dialogue compared to generic ChatGPT, and (3) fostered a deeper emotional connection and trust between the users and their conversational agents. The personas and dialogues in this figure were derived from our main study (Section 5).

[Ha et al., 2024]

This “Personality Traits” section includes the Big 5 (and other dimensions; some are close to the ones discovered in the work on the previous slides)

Figure 2: CloChat Design Lab Interface Features. The Design Lab interface comprises multiple pages, each linked to one of six categories from our literature review. Users input information into text fields for Demographic Cues (a) and Knowledge and Interest Cues (c). The Verbal Style Cues (b) page offers various language styles, selectable via checkboxes. Emoji options (d) are added through toggle switches. For the Appearance category (f), users describe the visual representation in text, detailed in Figure 4.

[Ha et al., 2024]



They use an LLM to first turn the user-provided traits into a descriptive persona text, which is then added to the user's request to prompt the LLM to respond to that request from that persona perspective.

That is, they are prompting the system to write/elaborate its own prompt.

Figure 3: Technical architecture of CloChat (Section 4.3). (Step 1) Given the non-visual traits from the CloChat design lab, we first convert them to a JSON specification (purple-filled box). (Step 2) We use GPT-4 to translate the JSON specification into a system message describing a persona (text with an orange background). (Step 3) We inject the system message into GPT-4, making it answer the user's message from the agent persona's perspective (text with a light-green background).

[Ha et al., 2024]

Critical reflection

On CUIs

Are CUIs/CAs really a “conversation”?

- **Conceptual critique:** We call this interaction „conversation“ but the **system is not actually a conversation partner**.
- “We reject the notion that such devices and interfaces are conversational in nature and that interaction with the interface is a conversation. Conversational Interaction [...] **confuses interaction with a device *within* conversation with an *actual* conversation**. [...] It is hard to make a case based on our data that responses from the device have a similar **status** to the conversation into which they are embedded.”
- **Example** of one such difference: A true conversationalist can turn a comment into a question by choosing to treat it as a question in their own response. This seems odd to users of CUIs/CAs – if you give Alexa an instruction, you might be unhappy with a response that treats it like a question (“Could you turn off the lights?” / “Yes, I could – are you asking me or telling me?”).



If you'd like to think further: Given this is from 2018, do you agree? Has sth. changed with more advanced (LLM) capabilities? If so, how?

[Porcheron et al., 2018]

“Unusual paper of the week” 😊

Analysing sci-fi speech interactions

“We study voice interactions with the *Enterprise*’s computer and compare them to current interactions.

Independent of futuristic computing power, we find key design-based differences:

Star Trek interactions are **brief** and **functional**, **not conversational**, they are highly **multimodal** and **context-driven**, and there is often **no spoken computer response**.”

[Axtell and Munteanu, 2021]



Tea, Earl Grey, Hot: Designing Speech Interactions from the Imagined Ideal of Star Trek

Benett Axtell

University of Toronto, TAGlab, Toronto, Canada

Cosmin Munteanu

University of Toronto Mississauga, ICCIT, Mississauga,

ABSTRACT

Speech is now common in daily interactions with our devices, thanks to voice user interfaces (VUIs) like Alexa. Despite their seeming ubiquity, designs often do not match users’ expectations. Science fiction, which is known to influence design of new technologies, has included VUIs for decades. *Star Trek: The Next Generation* is a prime example of how people envisioned ideal VUIs. Understanding how current VUIs live up to *Star Trek*’s utopian technologies reveals mismatches between current designs and user expectations, as informed by popular fiction. Combining conversational analysis and VUI user analysis, we study voice interactions with the *Enterprise*’s computer and compare them to current interactions. Independent of futuristic computing power, we find key design-based differences: *Star Trek* interactions are brief and functional, not conversational, they are highly multimodal and context-driven, and there is often no spoken computer response. From this, we suggest paths to better align VUIs with user expectations.

Video: <https://www.youtube.com/watch?v=DjGvY9Cigww>



Distraction-free interaction

Via voice – really?

Voice promises **hands-free use** (implying it might not distract you from your task) but it can still shift your focus away from your task!

- E.g. due to **bad error-handling**
(e.g. “Oh no, not that – how can I fix this? Come on, stop talking, that’s not what I wanted.”)
- E.g. demands **attention** and cognitive resources
(e.g. while driving, in the kitchen, ...)

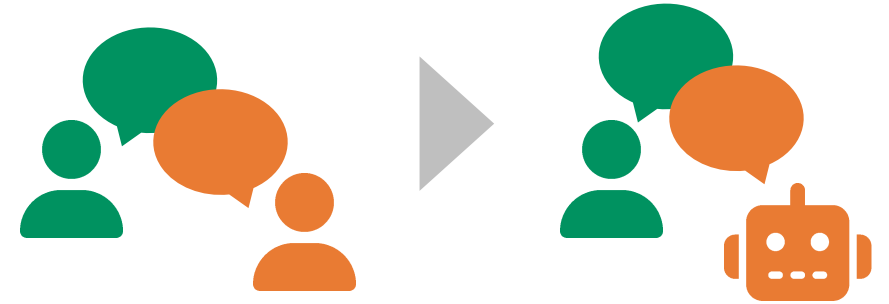


Photo by [C Technical](#)

Limitations of CUIs

Managing user expectations

- Users might **overestimate** machine capabilities
 - Normally we use voice / language to communicate with humans
 - Humans are intelligent
 - ...hence, the thing we talk to must intelligently think like a human
- Some **guidelines** for managing limitations:
 - Don't assume your CUI can fully understand the user's context
 - If in doubt, go for limited scenarios in the conversation
 - Make options and suggestions easily available to users
 - Make it explicitly clear it is not a human behind the text/voice



Also see people's prompting assumptions from the last session.

Outlook

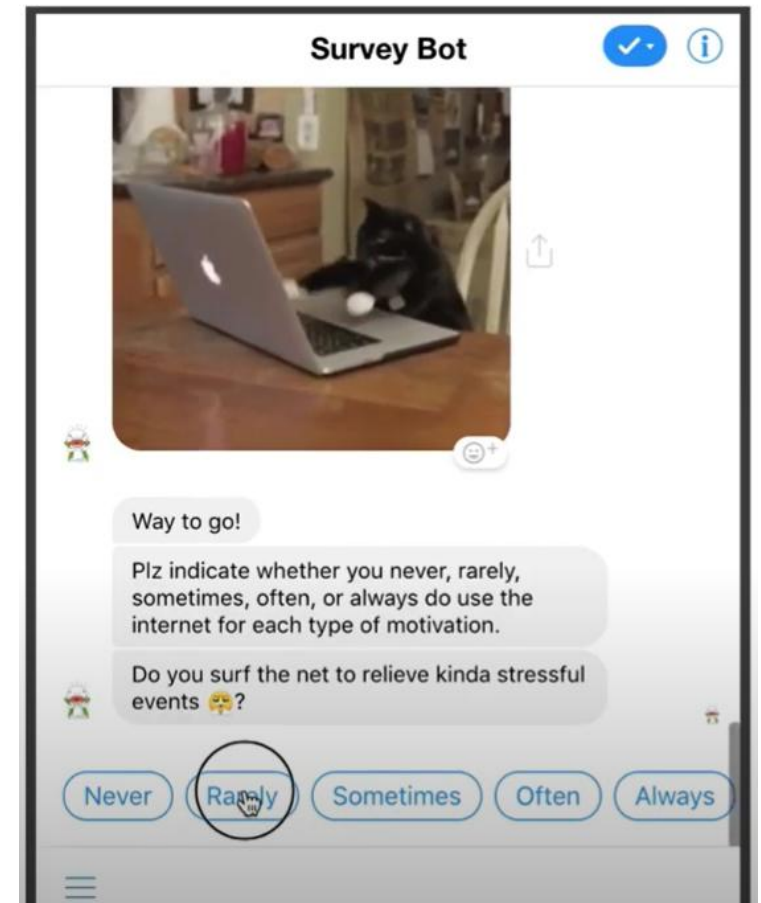
Chatbots as a research tool

Example: A chatbot as an interviewer

- Idea: Chatbot interviewer instead of a questionnaire
- Potential benefits? E.g. *“We found that the participants in the chatbot survey, as compared to those in the web survey, were more likely to produce differentiated responses and were less likely to satisfice*; the chatbot survey thus resulted in higher-quality data.”*
- Further recent work in this direction:
 - A Comparative Analysis of Information Gathering by Chatbots, Questionnaires, and Humans in Clinical Pre-Consultation ([link](#))
 - Chatbots for Data Collection in Surveys: A Comparison of Four Theory-Based Interview Probes ([link](#))

*satisfice - here: just answering to complete the questionnaire, not necessarily giving a „true“ response

[Kim et al., 2019]



<https://www.youtube.com/watch?v=A2AzbKFozc8>

Outlook

This is a very fast moving area of UI design but also very open still!

“Conversations with Documents” [Ter Hoeve et al., 2020]

The image shows the title page of the paper "Conversations with Documents: An Exploration of Document-Centered Assistance" by Maartje ter Hoeve, Robert Sim, Elnaz Nouri, Adam Fournery, Maarten de Rijke, and Ryan W. White. Below the paper is a diagram of the user interface. A "User" box asks, "Does the document already mention the mission of our company?". A "Digital assistant" box responds, "Yes, on page 2 it says: 'our mission is to increase people's productivity with the help of voice assistants.'". A "Document-centered assistance" box is also shown.

Copilot in Word [Microsoft, 2025]

The image shows the Microsoft Word interface with Copilot. The document is titled "Go-to-Market Document: Grand Opening of Contoso Home Improvement Store". The Copilot chat is open on the right, showing a conversation about the target audience. The chat includes a prompt guide, a response from Copilot, and a disclaimer: "AI-generated content may be incorrect".

Who is **person**

You can always use the prompt guide for suggestions by selecting this button

Who is our target audience

The target audience for the grand opening of the Contoso Home Improvement Store in Maricopa, Arizona encompasses three key segments: **homeowners and dedicated DIY enthusiasts, contractors and builders seeking quality supplies, and the local community at large.** By catering to these distinct groups, the store aims to establish itself as the ultimate hub for all home improvement needs in Maricopa. 1

Copy

AI-generated content may be incorrect

Catering to Contractors and Builders Competitive Analysis

Also see: https://en.wikipedia.org/wiki/Microsoft_Copilot

Talk to the Hand: an LLM-powered Chatbot with Visual Pointer as Proactive Companion for On-Screen Tasks

T. Prasongpongchai, P. Pataranutaporn, M. Lertsutthiwong, P. Maes

Integrating CUIs + GUIs

Designing for better integration by giving a chatbot a cursor

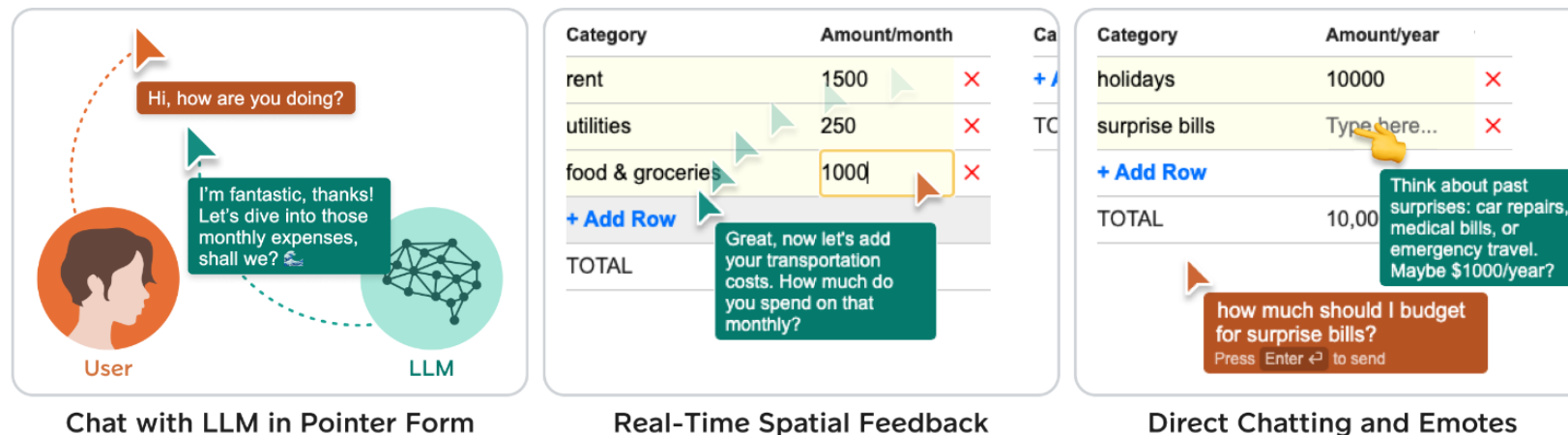


Figure 1: Example interactions of Pointer Assistant, an interaction technique for LLM-based human-AI interaction that features displaying the AI assistant as an animated extra mouse pointer (in green) alongside the user's pointer (in orange).

Q: Contrast this with other (currently popular) UI designs for chatbots.

- How are they usually integrated?
- E.g. when working on text, images, code?

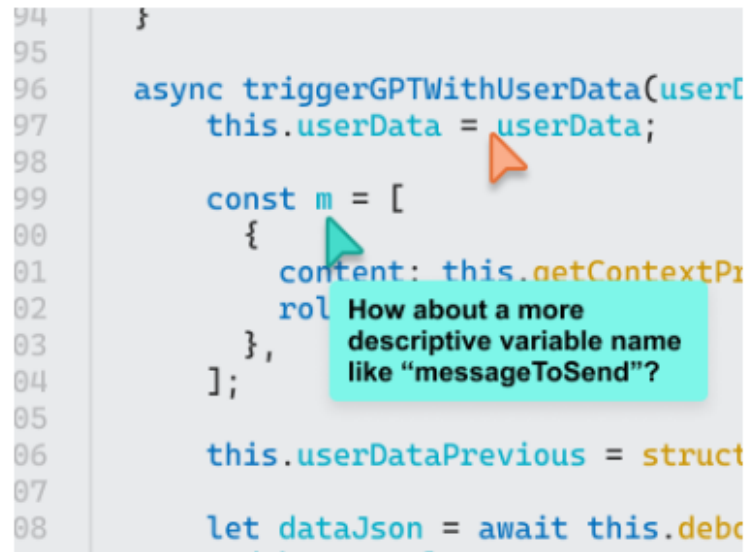
Key idea:

With other people, we can talk about something in the world. Can we bring a kind of **spatial referencing** to chatbots?

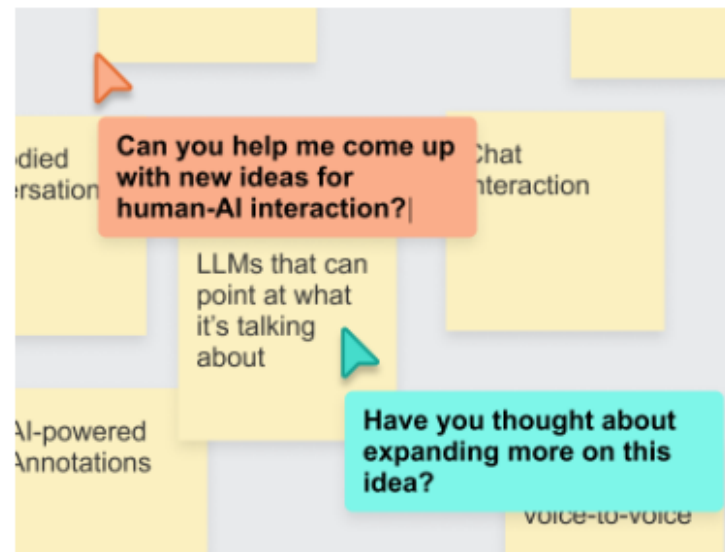
[Prasongpongchai et al., 2024]

More use-case ideas

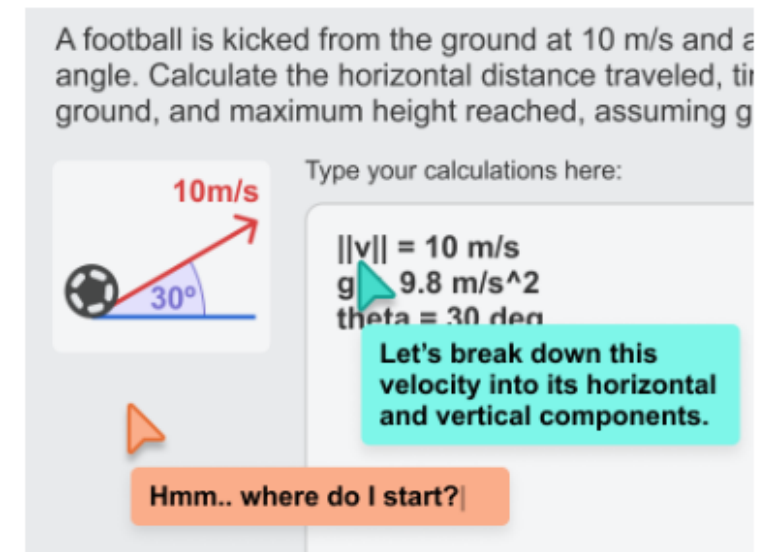
For the chatbot with a cursor



(a) Coding: Virtual Pair Programmer



(b) Brainstorming: Ideas Companion



(c) Education: Personal Tutor

Figure 2: Screen mock-ups for example potential use cases of Pointer Assistant in various domains.

[Prasongpongchai et al., 2024]

Implementation overview

For the chatbot with a cursor

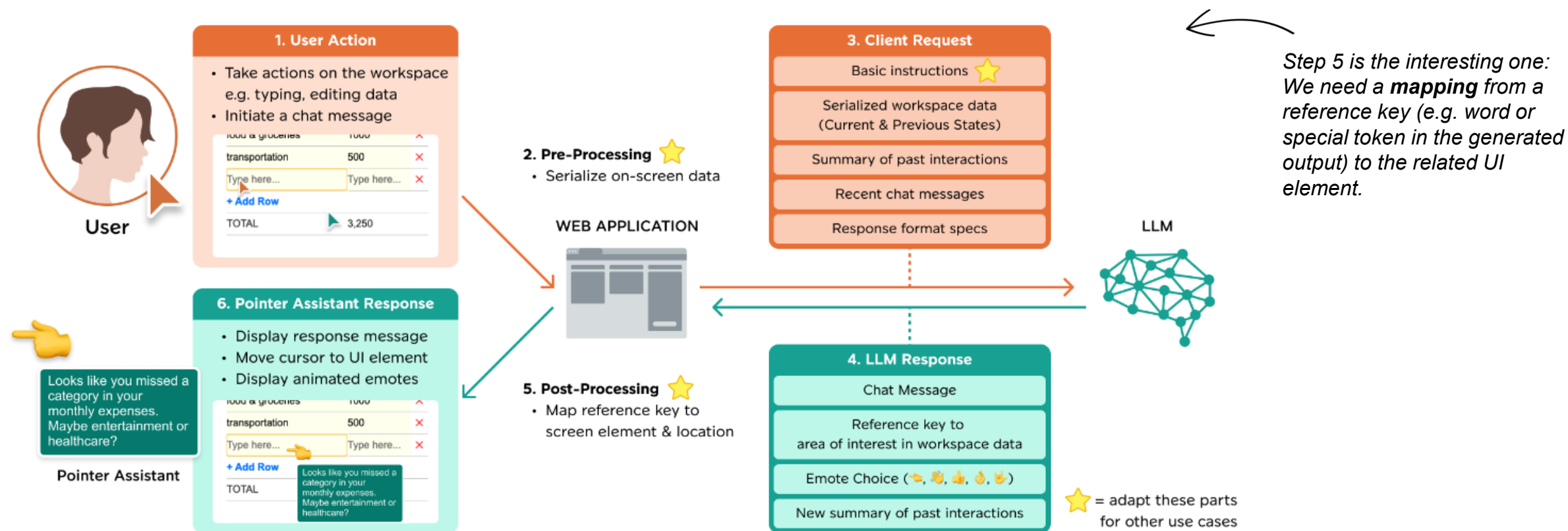


Figure 4: Interaction details including request and response message structures between the application and the LLM

[Prasongpongchai et al., 2024]

Outlook

Some research directions

- Many questions outdated / „solved“ now with LLMs? Really?
New questions, e.g. prompt-based conversation design?
- Also see last week (and the cursor example):
Integration of prompting and CUIs into GUIs?
- Developing theories of speech interface interaction:
e.g. *“a more robust theory of language production and interlocutor perceptions”*
- Work on speech interaction design:
e.g. *“what design methods and theories work best in a speech context”*
- Investigating multiple user contexts:
e.g. *“The study of situations that include multiple users and multiple speech interfaces [...]”*

[Clark et al., 2019]

References

References

- Mahla Alizadeh, Minha Lee, Dave Randall, Peter Tolmie, Dominik Pins, and Gunnar Stevens. 2026. Finding a Home for Voice Assistants: A Domestication Calculus Across Three Years and Thirty Households. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). Association for Computing Machinery, New York, NY, USA, Article 326, 1–19. <https://doi.org/10.1145/3772318.3791782>
- Ammari, T., Kaye, J., Tsai, J. Y., & Bentley, F. (2019). Music, Search, and IoT: How People (Really) Use Voice Assistants. *ACM Transactions on Computer-Human Interaction*, 26(3), 17:1-17:28. <https://doi.org/10.1145/3311956>
- Ashktorab, Z., Jain, M., Liao, Q. V., & Weisz, J. D. (2019). Resilient Chatbots: Repair Strategy Preferences for Conversational Breakdowns. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3290605.3300484>
- Bennett Axtell and Cosmin Munteanu. 2021. Tea, Earl Grey, Hot: Designing Speech Interactions from the Imagined Ideal of Star Trek. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 249, 1–14. <https://doi.org/10.1145/3411764.3445640>
- Choi, Y., Monserrat, T.-J. K. P., Park, J., Shin, H., Lee, N., & Kim, J. (2021). ProtoChat: Supporting the Conversation Design Process with Crowd Feedback. *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW3), 225:1-225:27. <https://doi.org/10.1145/3432924>
- Clark, L., Doyle, P., Garaialde, D., Gilmartin, E., Schlögl, S., Edlund, J., Aylett, M., Cabral, J., Munteanu, C., Edwards, J., & R Cowan, B. (2019). The State of Speech in HCI: Trends, Themes and Challenges. *Interacting with Computers*, 31(4), 349–371. <https://doi.org/10.1093/iwc/iwz016>
- Juhye Ha, Hyeon Jeon, Daeun Han, Jinwook Seo, and Changhoon Oh. 2024. CloChat: Understanding How People Customize, Interact, and Experience Personas in Large Language Models. In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 305, 1–24. <https://doi.org/10.1145/3613904.3642472>
- Kim, S., Lee, J., & Gweon, G. (2019). Comparing Data from Chatbot and Web Surveys: Effects of Platform and Conversational Style on Survey Response Quality. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3290605.3300316>
- McCrae, R. R., & Costa Jr, P. T. (2008). The five-factor theory of personality.

References

- Porcheron, M., Fischer, J. E., Reeves, S., & Sharples, S. (2018). Voice Interfaces in Everyday Life. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3173574.3174214>
- Reeves, B., & Nass, C. (1996). The media equation: How people treat computers, television, and new media like real people. Cambridge, UK, 10, 236605.
- Chek Tien Tan, Indriyati Atmosukarto, Budianto Tandianus, Songjia Shen, and Steven Wong. 2025. Exploring the Impact of Avatar Representations in AI Chatbot Tutors on Learning Experiences. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 1072, 1–12. <https://doi.org/10.1145/3706598.3713456>
- Ter Hoeve, M., Sim, R., Nouri, E., Fourney, A., Rijke, M. de, & White, R. W. (2020). Conversations with Documents: An Exploration of Document-Centered Assistance. *The 5th ACM SIGIR Conference on Human Information Interaction and Retrieval*. CHIIR. <https://doi.org/10.1145/3343413.3377971>
- Thanawit Prasongpongchai, Pat Pataranutaporn, Monchai Lertsutthiwong, and Pattie Maes. 2025. Talk to the Hand: an LLM-powered Chatbot with Visual Pointer as Proactive Companion for On-Screen Tasks. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 637, 1–16. <https://doi.org/10.1145/3706598.3715579>
- Sarah Theres Völkel, Ramona Schödel, Daniel Buschek, Clemens Stachl, Verena Winterhalter, Markus Bühner, and Heinrich Hussmann. 2020. Developing a Personality Model for Speech-based Conversational Agents Using the Psycholexical Approach. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376210>
- Sarah Theres Völkel, Samantha Meindl, and Heinrich Hussmann. 2021. Manipulating and Evaluating Levels of Personality Perceptions of Voice Assistants through Enactment-Based Dialogue Design. In *CUI 2021 - 3rd Conference on Conversational User Interfaces (CUI '21)*. Association for Computing Machinery, New York, NY, USA, Article 10, 1–12. <https://doi.org/10.1145/3469595.3469605>